



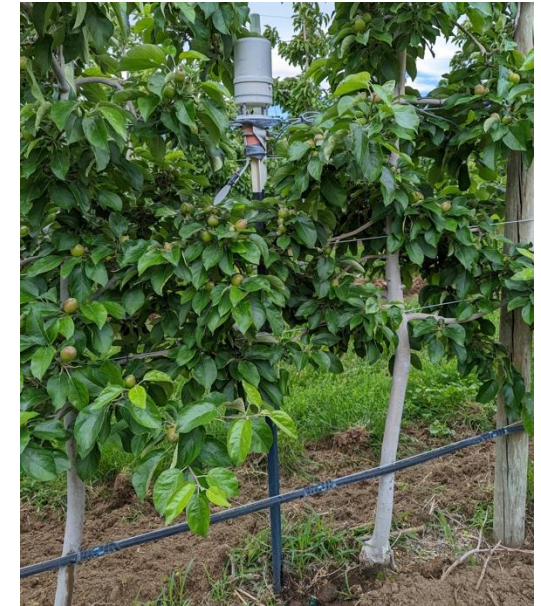
## Project Title: Modeling orchard effects on meteorological measurements

|        |              |                  |
|--------|--------------|------------------|
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Cooperators:

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# Objectives for Orchard Meteorology

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- 1. Quantify orchard effects on meteorological measurements**
- 2. Construct statistical models to predict in-orchard microclimate**
- 3. Develop and implement algorithms in AgWeatherNet**

# Key Findings

## 1. Consistent **growing season** orchard effects

- Peak offsets during May/June/July
  - Solar radiation ( $230 \text{ W m}^{-2}$ )
  - Lower air temperature ( $3.9 \text{ }^{\circ}\text{C}$ )
  - Relative humidity ( $- 32\%$ )
  - Wind speed ( $1.9 \text{ m s}^{-1}$ )

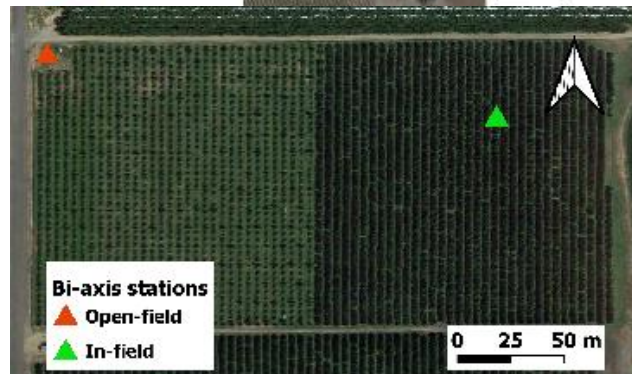
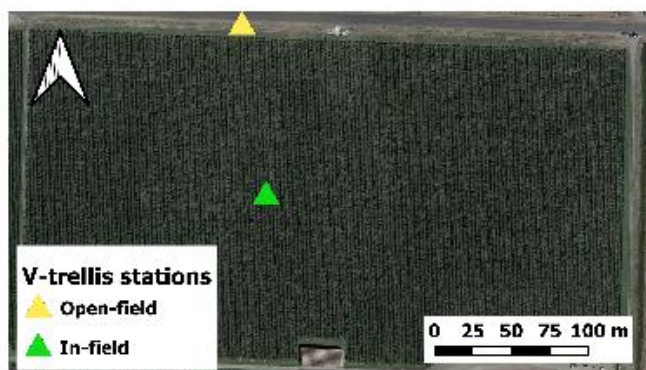
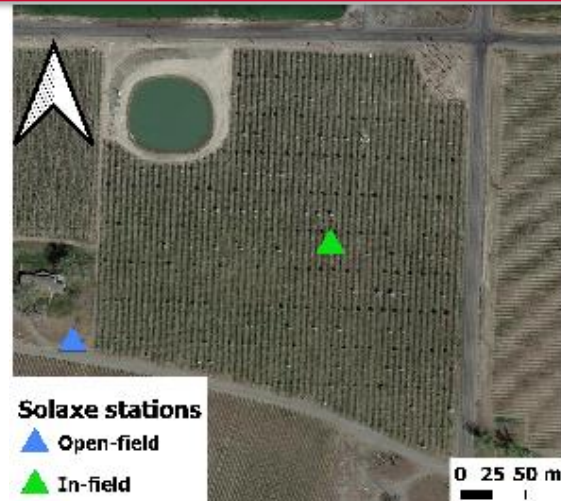
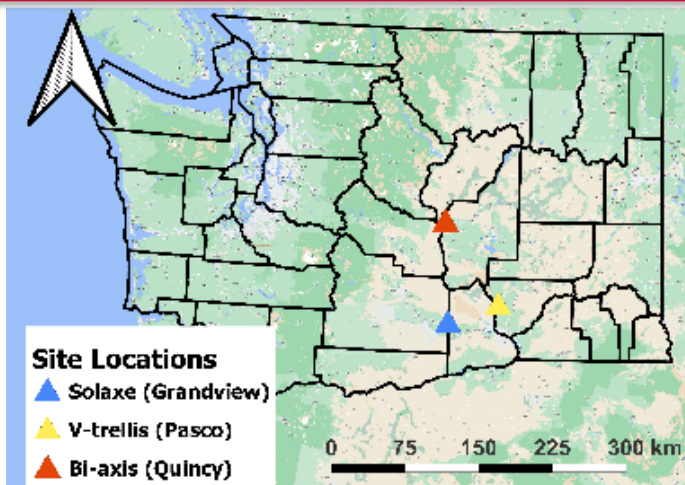
*Solaxe trellis apple orchard have the highest magnitude of orchard effects*

## 1. Seasonal hourly orchard effects

- *Timing of shading of the in-field weather station varies with different orchard training system*
- *Canopies trap warm air around noon and colder air overnight*

## 2. Machine Learning models can predict in-orchard microclimate

# Sites





# Training Systems



Solaxe

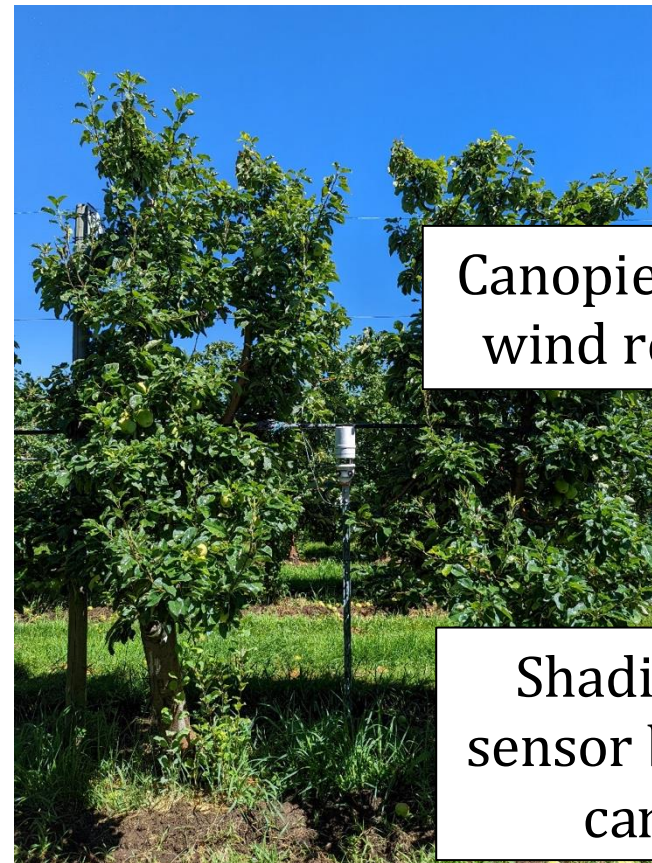
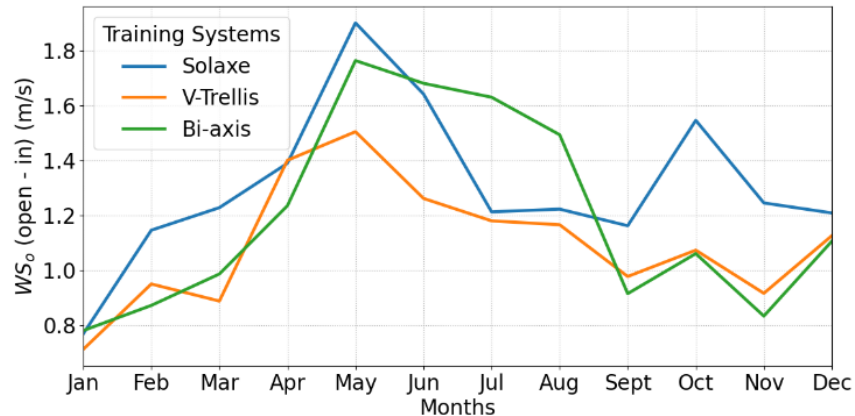
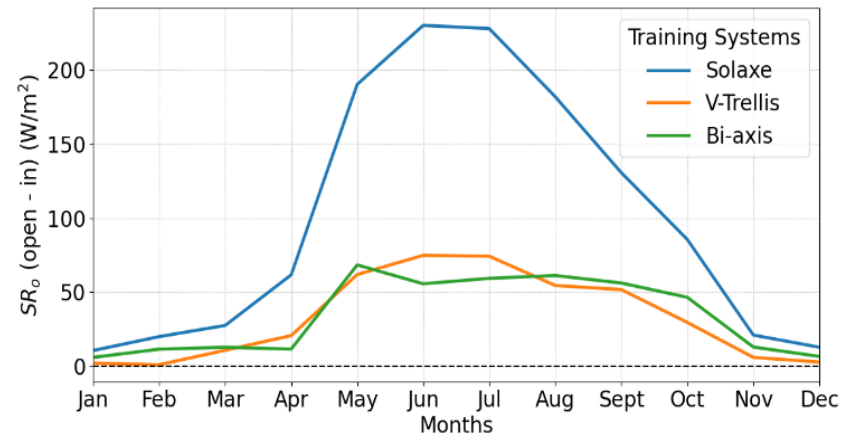


V-trellis



Bi-axis

# 2021-22 Monthly variation of Orchard Effects

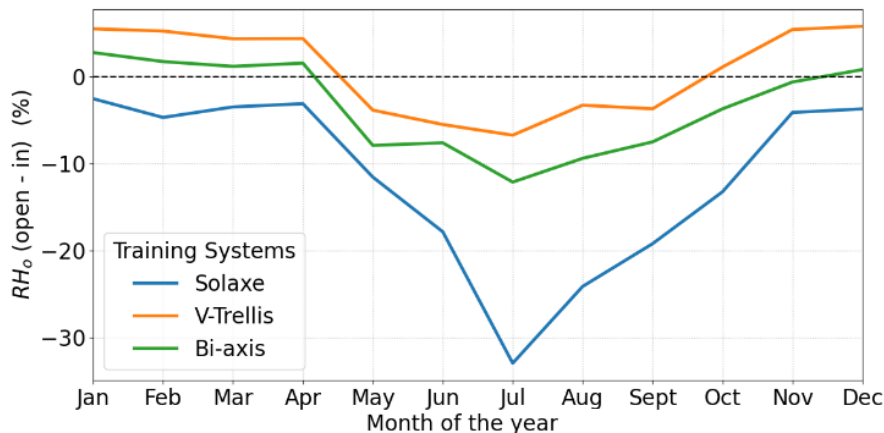
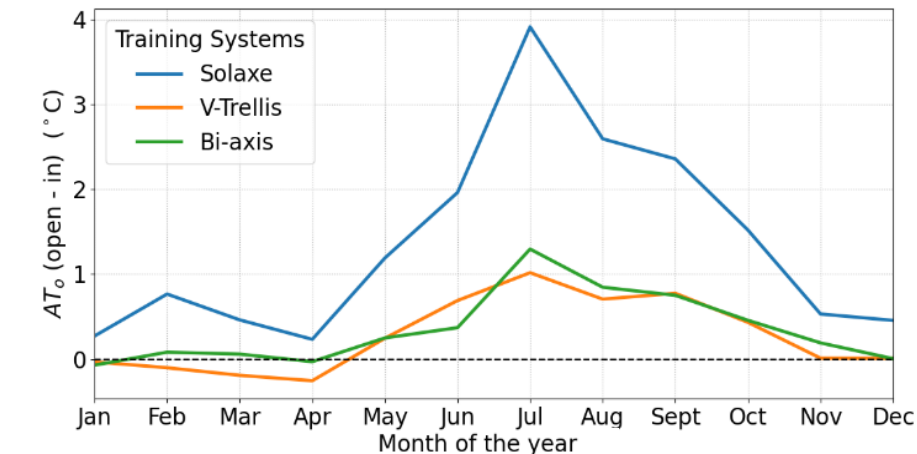


Canopies creating wind resistance

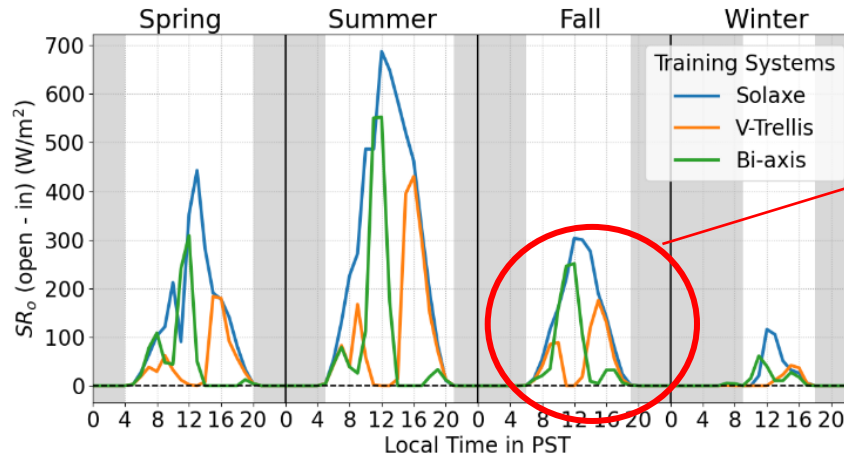
Shading of the sensor by orchard canopies



# 2021-22 Monthly variation of Orchard Effects



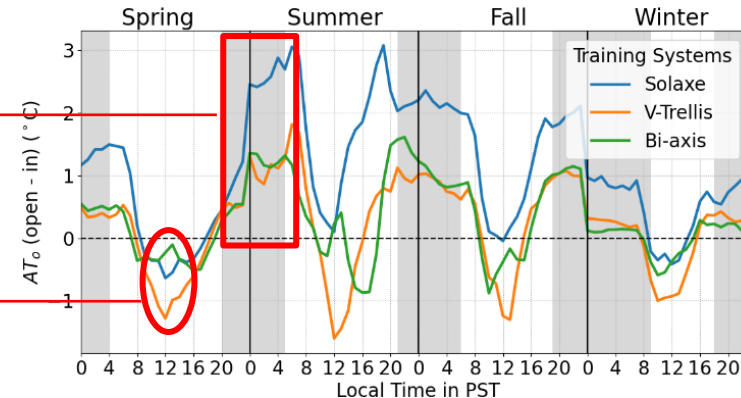
# 2021-22 Seasonal variation of Orchard Effects



Different timing of shading of sensors by the different Orchard training system

During the night, the orchards trap colder air

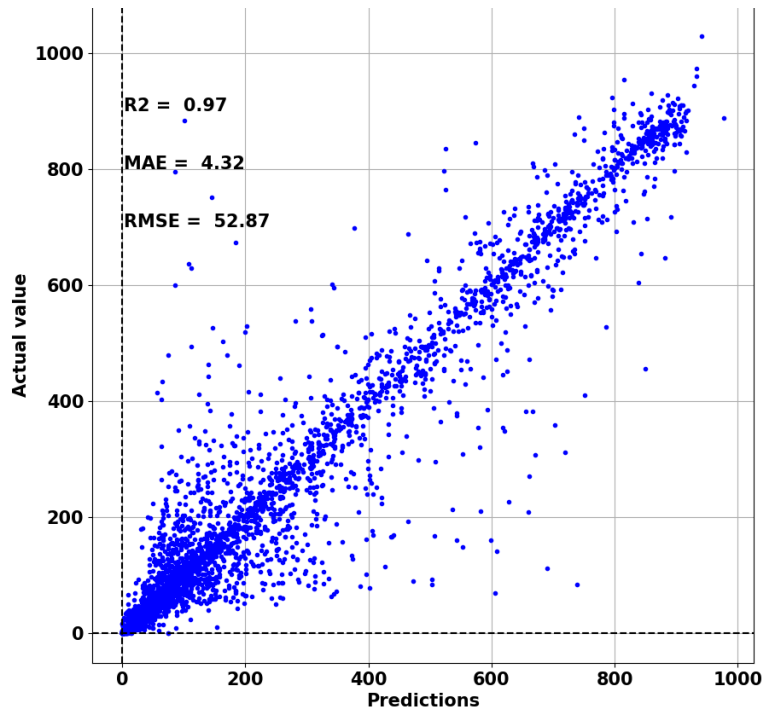
Orchard canopies trap warm air during daytime



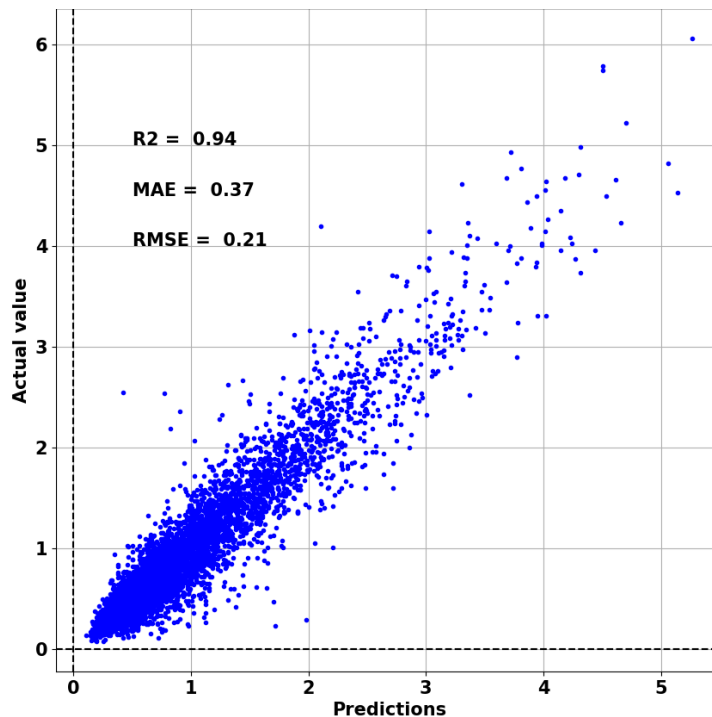


# Open-field to In-field Modeling

In-field **Solar radiation** prediction  
model ( $\text{W/m}^2$ )

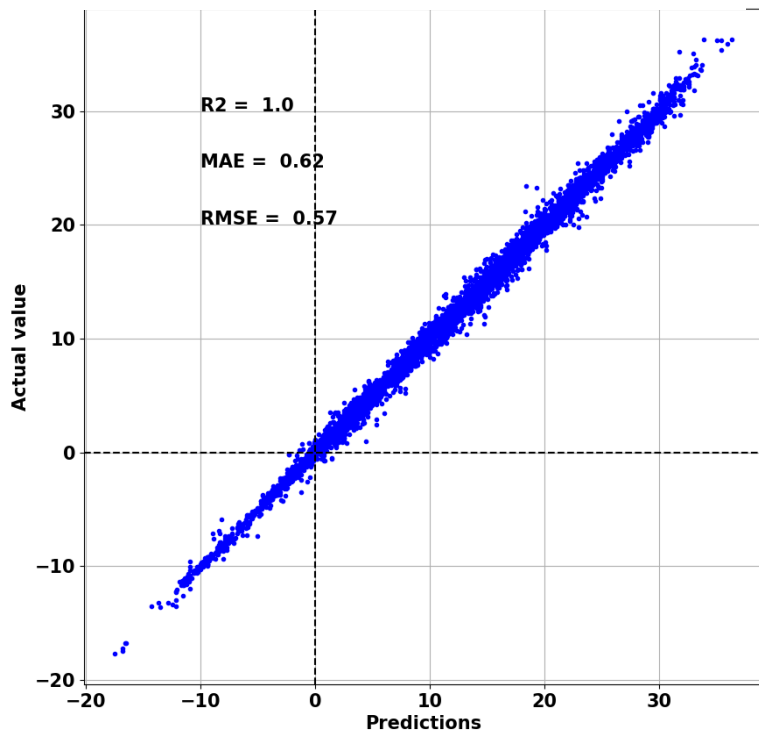


In-field **Wind Speed** prediction  
model ( $\text{m/s}$ )

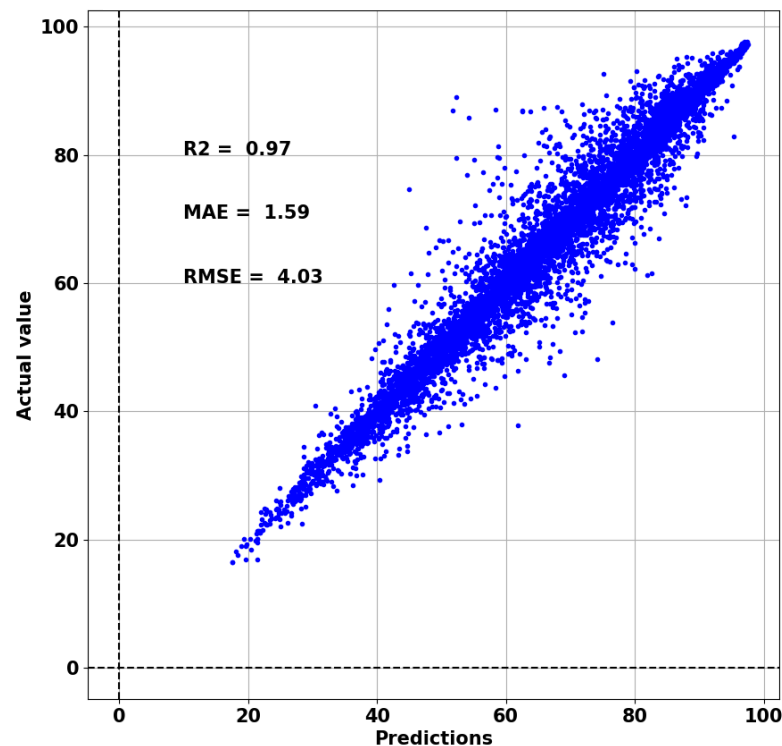


# Open-field to In-field Modeling

**In-field Air Temperature**  
prediction model (°C)



**In-field Relative Humidity**  
prediction model (%)





Questions?



Center for Precision &  
Automated Agricultural Systems



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